

A). Demo Purpose:

Condition	Pressure Range	Time Condition	Risk Score	Risk Status	LED	Buzzer	Clinical Interpretation
Normal Condition	< 32 mmHg	Any duration	0%	LOW	OFF	OFF	Normal pressure & microclimate
Moderate Risk	32–40 mmHg	< 20 sec	25%	MODERATE	ON	OFF	Early tissue stress possible
High Risk	40–60 mmHg	< 20 sec	50%	HIGH	ON	Slow Beep (~300 ms)	Reduced perfusion risk
Critical Risk (High Pressure)	> 60 mmHg	Immediate	100%	CRITICAL	ON	Fast Alert (~120 ms)	High bedsore formation risk
Critical Risk (Prolonged Pressure)	> 32 mmHg	> 20 sec	100%	CRITICAL	ON	Fast Alert (~120 ms)	Sustained pressure-induced tissue damage risk

B) Bedsores Condition :(Real code parameter)

Condition	Pressure Range	Time Condition	Risk Score	Risk Status	LED	Buzzer	Clinical Interpretation
Normal	< 32 mmHg	Any / brief	0%	LOW	OFF	OFF	Below capillary-closure reference; monitor for shear and moisture.
Moderate	32–40 mmHg	15–30 min	25%	MODERATE	ON	OFF	Early loading stress; risk starts increasing.
High	40–60 mmHg	5–15 min	50%	HIGH	ON	Slow Beep	Higher ischemic risk; faster warning needed.
Critical (High Pressure)	> 60 mmHg	Immediate	100%	CRITICAL	ON	Fast Alert	Very high loading; quick perfusion compromise.
Critical (Prolonged)	> 32 mmHg	> 1–4 hours	100%	CRITICAL	ON	Fast Alert	Sustained pressure strongly linked to pressure-ulcer development.